The fusion of the oldest activity (agriculture) with the latest technology (drones)

Yield estimation. "Estimating yield through drone-imaging solutions is one of the top use-cases in India. Drones can be used to carry out yield estimations not just across different patches of lands but on different types of crops as well. Drones can be used to assess yield on almost every single crop type," says Mughilan Thiru Ramasamy, co-founder, Skylark Drones.

Levelling lands. Drones can be used to gauge the unlevelled stretch of agricultural lands, and then the same data can be transferred to machines that are used to level the land pieces.

Irrigation. "On the irrigation side, it is all about using the right elevation and pressure to ensure the maximum growth of crops. With our drone solutions, we have been able to save five times the effort with fifteen per cent improvement in cost-efficiency," points out Vipul Singh, CEO & founder, Aarav Unmanned Systems.

Crop insurance. "During floods and overcast conditions, drones can be flown at very low altitudes to obtain high-quality images of the ground. This is something that can help the farmers and insurance companies alike," as per Vipul Singh.

Drones are also being used to monitor crop health, spraying pesticides as well as for surveillance of agricultural lands throughout India.

when they collaborate, they will be able to produce better farm incomes and also adopt best agritech practices and innovations," says Praveen Kumar of Unnmiti.

He adds, "What smart farming does is that it enables a farmer to answer natural challenges in a very efficient manner. With the kind of data that are now being generated by different farms, the day is not far away when farmers will be able to gauge the impact of their decisions even before they take them."

Drones in agritech: The biggest example

Startups using drones for improving agricultural practices is probably the biggest example of the agritech revolution getting brewed in the country. From being used right from the first step of levelling land pieces and sowing seeds, drones have found usage in almost every step that encompasses a farmer's life.

The next generation of farmers, as Ashish Kumar from Amosta explains, is open to experimenting with new technologies in agriculture. He says, "Another thing is that not every farmer's attitude will shift to using agritech and drones."

"As far as ROI is concerned, there are farmers with the mindset that if a work can be done with manual labour, why should they shift to using technology like drones? Now the interesting part is that a lot of these farmers now use tractors instead of manual labour, and when tractors were first introduced, they were not adopted immediately. But today, a lot of them use tractors," he says.

Big corporate houses like Pepsico, farmer's cooperative societies, and big landowners, as per Ashish, will be the first ones to use drones and other mediums of agritech. He noted that these will be the ones driving the ROI in the first adoption phase, and as technology in drones becomes economical to use, small-scale farmers will start adopting them, driving the ROI further.

The academia is playing its role too

The focus of a lot of students from the different engineering domains, as per Geetali Saha, associate professor, G.H. Patel College of Engineering and Technology, has been on taking up projects around agritech. She points out that the young generation of India understands the value of agriculture and is inclined towards making the conditions better in the country.

Engineering students taking interest in building solutions for farmers, as Professor Saha informs, is a sign of India's bright future in the agritech stage. She explains that the earlier the students start working on a technology, the better solutions they can tailor.

"Students taking up agritech projects are not only good for the country but also for the budding entrepreneurs. These projects give students an edge when it comes to starting their entrepreneurial journey," professor Saha says.

Big question: Can India lead?

Given the arguments made at several Agritech panel discussions held at various editions of India Technology Week, it is clear that India has a golden opportunity to lead in the technology for agriculture space, and drones being used in various applications of agriculture is one of the biggest proofs of the same. The government of India, recently notifying that it will be funding 112 agritech startups in the country, is another hint towards the long-term plans of the country.

Union Minister for Agriculture Narendra Tomar, in July 2020, recalling Prime Minister Narendra Modi's words, had also reiterated promotion of startups and agri-entrepreneurs. He had said that the government is working to ensure innovation and use of technology in agriculture and allied sectors.

The fact that a lot of countries, just like India, are constituted of small-land farmers is another proof that what we develop in India can be easily adopted and used by these countries. Given India's prowess in developing low-cost yet top-of-the-mark technological solutions is also a clear hint that, if given economical agritech solutions, even bigger landowners from economies like the United States will not hesitate in adopting these solutions.

Yes, early adopters in the technology adoption curve define whether a technology will be a hit or a miss, but it is nowhere written that these early adopters have to be from developed countries. Countries falling in the continent of Africa, Europe as well as countries in South America, depend a lot on agriculture and, apart from having a lot of small land farmers, these countries are also blessed with climatic conditions very similar to India. As a matter of fact, it would be better to say that India is blessed with almost every type of climate present in different parts of the world, a requisite for defining agricultural activities! EFY